

Toward Standardized Data Sets for Climate Model Experimentation

A new initiative collects, archives, and documents climate forcing data sets to support coordinated modeling activities that study past, present, and future climates.

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Climate change models simulate historical changes and predict future trends in a variety of phenomena, including land cover types like these rain forests and coastal mountain ranges in South America. A new initiative, input data sets for Model Intercomparison Projects (input4MIPs), assists climate modelers by collecting, documenting, and archiving the forcing data sets used to create these models. Credit:Petmal/iStock/Getty Images Plus

Climate models are the most powerful tools for investigating global climate changes caused by human activity. These models are based on mathematical representations of the physics, chemistry, and biology of the climate system. The ways these models respond to externally imposed conditions, often representing the past or a hypothesized future, help scientists understand the forces and processes responsible for long-term changes in Earth's climate.

Although climate models have improved over time, no single model perfectly represents all aspects of the climate system. Thus, rather than depending on the veracity of a single model, scientists rely on simulations by multiple models, each with its own strengths and limitations, to provide a range of likely outcomes. Each new model version revisits key standard experiments, providing scientists with a perspective concerning the robustness of results. The key to this strategy is that each model is “forced” by the same set of externally imposed conditions.

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The [World Climate Research Programme \(WCRP\)](#) [Coupled Model Intercomparison Project \(CMIP\)](#) has led the effort to coordinate international climate model experimentation. A critical element in the CMIP protocol is determining what forcing data sets are most suitable for each experiment. In the past, it was not essential to formally document every aspect of every simulation because model simulations were mainly of interest to the climate modeling groups themselves; modelers were well aware of the experiment design and the forcing applied. Now, however, future climate projections from CMIP models may form part of the

basis for high-stakes decisions, so it is imperative to strictly and formally document experimental conditions, including the “forcing” data used to drive simulations.

Phase 6 of the CMIP project ([CMIP6](#)) is currently under way, with the first data expected in mid- to late 2018. For the first time, the forcing data sets used in generating climate simulations of the past, present, and future will be collected, centrally archived, and documented under a new initiative known as input data sets for Model Intercomparison Projects ([input4MIPs](#)). Better provenance and comprehensive documentation provided by this initiative will improve consistency, reproducibility, and traceability of the suite of CMIP6 model results.

Charge of input4MIPs

In assessing the limits of our understanding of the climate system, scientists have exploited the results of CMIP and its predecessors. [As far back as 1990](#), multimodel experimentation provided the basis for many of the conclusions of the [Intergovernmental Panel on Climate Change assessment reports](#).

The first phase of CMIP was [initiated in 1995](#). Along with the original Atmospheric Model Intercomparison Project ([AMIP](#)), CMIP inspired a profusion of other MIPs. The expansion of intercomparison activities underlies a new imperative in the most recent CMIP phase, [CMIP6](#), namely, to impose [tighter coordination](#) and more [uniform reliance](#) on a common, [community-wide](#) infrastructure.

In support of CMIP, the new input4MIPs initiative promises to improve transparency, documentation, and consistency in the forcing data sets that are imposed in climate simulations. Some of these data (Figure 1) are based on observations (e.g., sea surface temperatures or [greenhouse gas](#) concentrations needed to force simulations of historical and future climate change). Others, like the forcing data employed in the projections called for by [ScenarioMIP](#), are based on hypothetical future conditions. The input4MIPs data products have been endorsed for CMIP6 use and are considered the best available for generating simulations of past, present, and future climate.

Meeting the Challenge

CMIP forcing data cover a broad range of prescribed quantities. Examples include greenhouse gas concentrations (Figure 2) or emissions, variability of the solar “constant,” and [land use](#) changes (Figure 3). For atmosphere-only simulations (AMIP), data include sea surface temperatures and sea ice concentrations.

Until now, the data sets prepared by experts from different scientific subdisciplines have not conformed to uniform standards; the data were hosted on different servers in whatever structure and format seemed convenient. This diversity impaired interpretation and required special coding for use. Some forcing data sets lacked version control and were poorly curated.

The CMIP experiment design requires that the same external conditions be specified across all models and that that these forcing data sets be well documented. Differences in model responses can then be attributed solely to differences in the model formulation (discounting uncertainty associated with climate noise).

To ensure that the same forcing data are applied in all CMIP6 simulations, input4MIPs now maintains the official forcing data set collection. [Standards](#) have been imposed, ensuring that all data are similarly structured and include metadata describing their contents, structure, and domain. These standards follow the CMIP6 model output conventions and, where possible, adopt the CMIP-mandated names for variables and attributes.

A [special CMIP6 issue](#) of *Geoscientific Model Development* documents each of the new forcing data sets, and an up-to-date summary of input4MIPs data is maintained on the [input4MIPs project website](#). As the project matures, digital object identifiers ([DOIs](#)) will make each [data set citable](#). These DOIs ensure that [data provenance](#) can be traced and data creators can be credited. All officially released contributed data versions will be archived by input4MIPs and remain accessible in perpetuity.

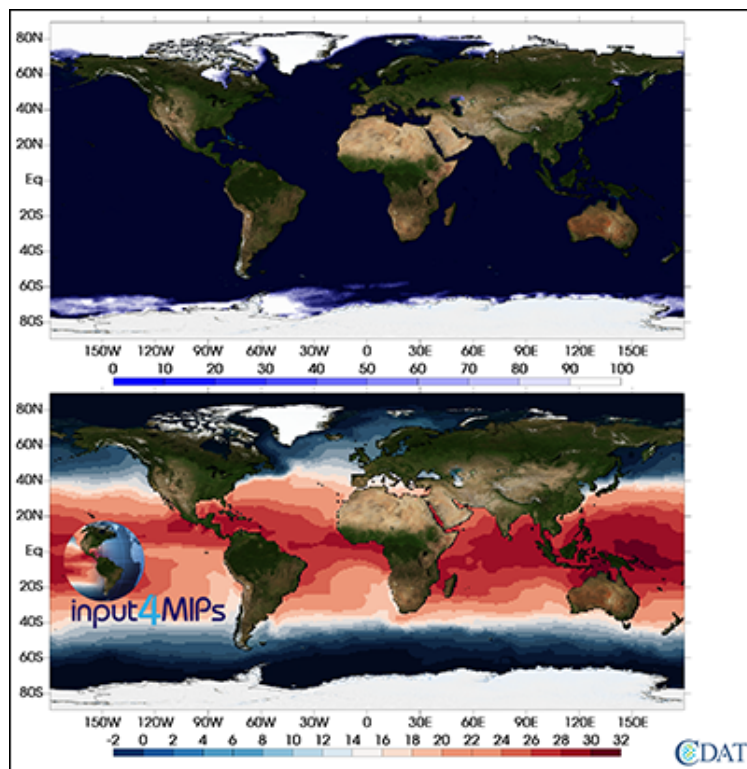


Fig. 1. The history of sea surface temperature and sea ice from 1870 to the present is used as a boundary condition for the atmospheric-only simulations that contribute to the Atmospheric Model Intercomparison Project (AMIP) experiment, which dates back to 1990. [This experiment](#), the longest-running MIP, is part of the CMIP6 DECK. Shown here are conditions during December 2017 for (top) the mean sea ice concentrations (scale shows percentage of each cell's area covered by ice), captured from the observations used in PCMDI AMIP boundary conditions (PCMDI-AMIP-1-1-4), and (bottom) sea surface temperature (scale shows degrees Celsius). The figure was created using Community Data Analysis Tools ([CDAT](#)). The data set was provided by the Program for Climate Model Diagnosis and Intercomparison.

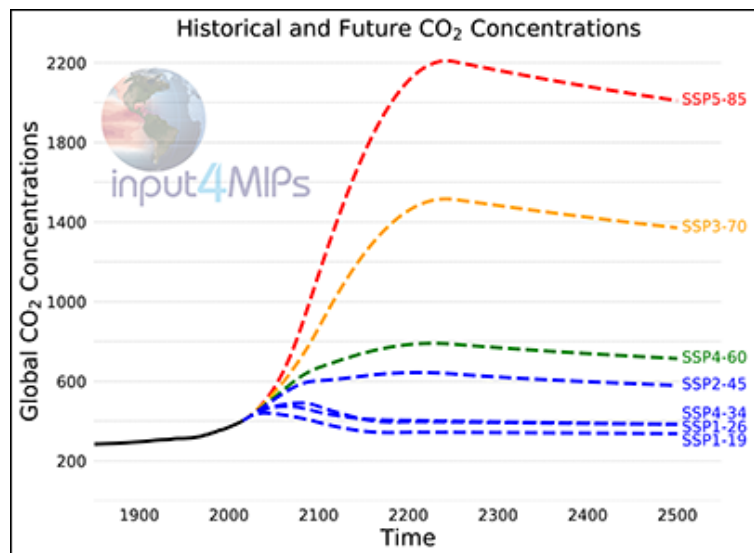


Fig. 2. CMIP6 data for historical carbon dioxide (CO₂) emissions for the period extending through 2014, followed by seven shared socioeconomic pathways (SSPs) that cover a broad range of possible future scenarios for greenhouse gas emissions, extending to the year 2500. Scenario SSP5-85 assumes fossil-fueled development, which yields an effective radiative forcing of 8.5 watts per meter. Scenario SSP1-19 assumes the lowest emissions and predicts negative CO₂ emissions growth around 2050. The data set was provided by the Australian-German Climate and Energy College, University of Melbourne, Australia.

current input4MIPs holdings.

Input4MIPs Holdings

Before a model is used in scientific studies, CMIP requires that output from a small set of well-established baseline experiments be made available for community scrutiny. Performing the CMIP DECK experiments and the CMIP historical simulation establishes a model's credentials, and any improvements in [model performance](#) can be objectively determined. All data sets needed for the DECK and historical simulations are now available.

Building on the DECK and historical experiments, CMIP6 has endorsed additional MIPs that focus on gaps in our understanding of the climate system. The [23 CMIP6-Endorsed MIPs](#) call for [285 climate experiments](#), and many of these require additional forcing data sets. Input4MIPs will, with some exceptions, collect these data sets, some of which are already available (Table 1, second section).

Managing Changes

Because input4MIPs is a research-based project, official releases of data might be flawed and require correction. Some data will be updated periodically and extended as new data become available. Input4MIPs has implemented strict version control of the data releases, which aids transparency and traceability.

The [Earth System Grid Federation](#) (ESGF) hosts and serves input4MIPs data alongside the CMIP data. The CMIP Panel released the initial CMIP6 Forcing Data sets for the CMIP Diagnosis, Evaluation, and Characterization of Klima (DECK) and historical simulations version 6.0 for use on 20 December 2016, and two subsequent versions have superseded this release: version 6.1.1 on 22 May 2017 and version 6.2 on 11 September 2017, as documented on the [input4MIPs website](#). Table 1 provides a summary of

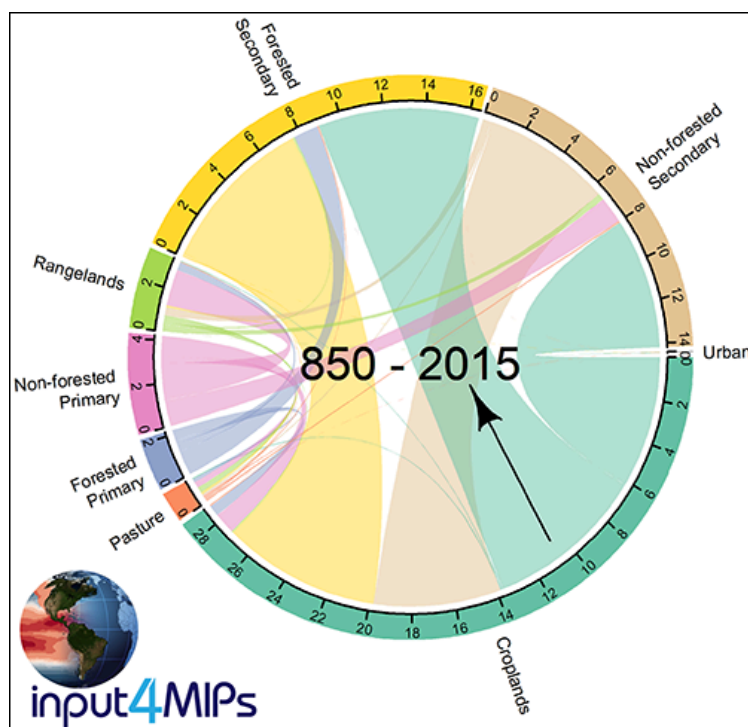


Fig. 3. This land use harmonization data set (LUH2, UofMD-landState-2-1-h) provides a harmonized set of land use scenarios that smoothly connects the changes of land use over the historical period from 850 to 2015. Colored arcs represent the annual changes between land use states. The color of each arc represents the land use category from which transition occurs to a different category, and the width of the arc represents the annual land use area undergoing the transition (e.g., the wide yellow arc shows that nearly 75,000 square kilometers of croplands transition to forested secondary lands every year). The data set was provided by the University of Maryland.

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Many of the input4MIPs data sets build on those used in previous CMIP phases. Some have been revised to include newly acquired observational estimates and to be extended at least to the end of 2014; others have been expanded to include forcing needed by newer, more comprehensive models, like those for the carbon and [nitrogen cycles](#) (see Table 1 for details).

Ongoing Contributions and Future Evolution

The input4MIPs initiative represents the latest addition to the U.S. Department of Energy’s 30-year scientific leadership through the Program for Climate Model Diagnosis and Intercomparison (PCMDI). This program’s other efforts include [scientific leadership](#) of CMIP experiment design, development of [climate and forecast conventions](#) and data [standards for CMIP](#), [obs4MIPs](#), [archiving of data from earlier CMIP phases](#), and [ESGF development and operations](#). These efforts have fostered collaboration across more than 27 countries involved in the [WCRP CMIP project](#).

With time, input4MIPs will include whatever new or revised data are needed in subsequent phases of CMIP. As the project matures and participants adhere more closely to the input4MIPs data specifications, the publication process will incorporate additional data set quality control checks. We also plan to partner with modeling groups to evaluate the impact on simulations of each new forcing data set prior to its acceptance for use in future CMIP phases.

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**Table 1. Forcing Data Sets (Version 6.2.11)
Generated for Use in the CMIP6 DECK,
Historical, and Satellite MIP Experiments****CMIP6 DECK and Historical Simulations**

Data Set	Temporal Range	Data Set Version	Documentation
Historical anthropogenic short-lived climate forcing (SLCF), carbon dioxide and methane emissions	1750–2014	2017-05-18, 2017-08-30, 2017-10-05 (aircraft sulfur dioxide)	<i>Hoesly et al.</i> [2017a, 2017b, 2017c, 2017d, 2018]
Historical biomass burning emissions	1750–2015	1.2	<i>van Marle et al.</i> [2016, 2017]
Historical land use changes	850–2015	2.1h	Land-Use Harmonization (LUH2), <i>Hurt et al.</i> [2017]
Historical greenhouse gas historical concentrations	0–2014	1.2.0	<i>Meinshausen and Vogel</i> [2016], <i>Meinshausen et al.</i> [2017]
Historical stratospheric aerosols	1850–2014	3.0.0	<i>Thomason et al.</i> [2017]
Historical ozone	1850–2014	1.0	<i>Hegglin et al.</i> [2016a]
Historical nitrogen deposition	1850–2014	2.0	<i>Hegglin et al.</i> [2016b]
Solar forcing	1850–2299	3.2	<i>Matthes et al.</i> [2017a, 2017b]
Aerosol optical properties	1850–2100	1.0	<i>Stevens et al.</i> [2017]
Historical sea surface temperature (SST) and sea ice	1870–2017	1.1.4	<i>Durack and Taylor</i> [2018]

CMIP6-Endorsed MIP Forcing Data Sets

MIP/Data Set	Temporal Range	Data Set Version	Documentation
CFMIP: cloud feedbacks		1.0	<i>Webb et al.</i> [2017]
C4MIP/OMIP: carbon isotopic	1850–2015	1.1, 2.0	<i>Jones et al.</i> [2016], <i>Orr et al.</i> [2017], <i>Graven et al.</i> [2017a, 2017b]
DAMIP: ozone	1850–2014	1.0, 1.1	M. I. Hegglin et al. (database in preparation), <i>Plummer et al.</i> [2017, 2018a, 2018b, 2018c]
DCPP: idealized SST restoring fields	1900–2013	1.1	<i>Boer et al.</i> [2016], <i>Cassou et al.</i> [2017a, 2017b]
FAFMIP: heat and freshwater forcing fields	Idealized future	2.1.0	<i>Gregory et al.</i> [2016], <i>Gregory</i> [2018]
HighResMIP: SST and sea ice boundary conditions	1948–2014	2.2.0.0-r0	<i>Haarsma et al.</i> [2016], <i>Kennedy et al.</i> [2017]

OMIP (JRA55-do): ocean boundary conditions	1958–2018	1.3	<i>Griffies et al.</i> [2016], <i>Orr et al.</i> [2017], <i>Tsujino et al.</i> [2018], <i>Graven et al.</i> [2017a, 2017b]
RFMIP: atmospheric conditions for offline radiative transfer calculations		0.4	<i>Pincus et al.</i> [2016], <i>Pincus</i> [2017]
ScenarioMIP (several data sets are pending) ^a	2015–2100 or 2500; TBD	2.1f; 1.2.0; TBD	LUH2 ; <i>Meinshausen et al.</i> [2017]; TBD

^aTBD = to be determined.

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